

gilmore-lie-groups: table evidence

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_ TAB_001	26	0.877	$\begin{tabular}{t}{ l l l l l }\hline & I & (123) & (321) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & v_1 = s_1 + s_2 + s_3 \\ \Gamma^2 & 1 & \omega & \omega^2 & v_2 = s_1 + \omega s_2 + \omega^2 s_3 \\ \Gamma^3 & 1 & \omega^2 & \omega & v_3 = s_1 + \omega^2 s_2 + \omega s_3 \\ \hline\end{tabular}$	$\begin{array}{c ccc} & I & (123) & (321) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & v_1 = s_1 + s_2 + s_3 \\ \Gamma^2 & 1 & \omega & \omega^2 & v_2 = s_1 + \omega s_2 + \omega^2 s_3 \\ \Gamma^3 & 1 & \omega^2 & \omega & v_3 = s_1 + \omega^2 s_2 + \omega s_3 \\ \hline\end{array}$	$\begin{array}{c ccc} & I & (123) & (321) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & v_1 = s_1 + s_2 + s_3 \\ \Gamma^2 & 1 & \omega & \omega^2 & v_2 = s_1 + \omega s_2 + \omega^2 s_3 \\ \Gamma^3 & 1 & \omega^2 & \omega & v_3 = s_1 + \omega^2 s_2 + \omega s_3 \\ \hline\end{array}$
gilmore-lie-groups_ TAB_002	30	0.669	$\begin{tabular}{t}{ l l l l l l }\hline & (I) & (12)(34) & (13)(24) & (14)(23) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & 1 & w_1 = t_1 + t_2 + t_3 + t_4 \\ \Gamma^2 & 1 & 1 & -1 & -1 & w_2 = t_1 + t_2 - t_3 - t_4 \\ \Gamma^3 & 1 & -1 & 1 & -1 & w_3 = t_1 - t_2 + t_3 - t_4 \\ \Gamma^4 & 1 & -1 & -1 & 1 & w_4 = t_1 - t_2 - t_3 + t_4 \\ \hline\end{tabular}$	$\begin{array}{c ccccc} & I & (12)(34) & (13)(24) & (14)(23) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & 1 & w_1 = t_1 + t_2 + t_3 + t_4 \\ \Gamma^2 & 1 & 1 & -1 & -1 & w_2 = t_1 + t_2 - t_3 - t_4 \\ \Gamma^3 & 1 & -1 & 1 & -1 & w_3 = t_1 - t_2 + t_3 - t_4 \\ \Gamma^4 & 1 & -1 & -1 & 1 & w_4 = t_1 - t_2 - t_3 + t_4 \\ \hline\end{array}$	$\begin{array}{c ccccc} & I & (12)(34) & (13)(24) & (14)(23) & \text{Basis functions} \\ \hline \Gamma^1 & 1 & 1 & 1 & 1 & w_1 = t_1 + t_2 + t_3 + t_4 \\ \Gamma^2 & 1 & 1 & -1 & -1 & w_2 = t_1 + t_2 - t_3 - t_4 \\ \Gamma^3 & 1 & -1 & 1 & -1 & w_3 = t_1 - t_2 + t_3 - t_4 \\ \Gamma^4 & 1 & -1 & -1 & 1 & w_4 = t_1 - t_2 - t_3 + t_4 \\ \hline\end{array}$
gilmore-lie-groups_ TAB_003	62	0.837	$\begin{tabular}{t}{ l l }\hline & \text{Group} & \text{Dimension} \\ \hline UT(p, q) & p^2 + q^2 + pq \\ HT(p, q) & p(p + q) \\ UT(p, q, r) & p^2 + q^2 + r^2 + pq + pr + qr \\ Sol(n) & n(n + 1)/2 \\ Nil(n) & n(n - 1)/2 \\ A(p, q) & pq \\ \hline\end{tabular}$	$\begin{array}{c c} \text{Group} & \text{Dimension} \\ \hline UT(p, q) & p^2 + q^2 + pq \\ HT(p, q) & p(p + q) \\ UT(p, q, r) & p^2 + q^2 + r^2 + pq + pr + qr \\ Sol(n) & n(n + 1)/2 \\ Nil(n) & n(n - 1)/2 \\ A(p, q) & pq \\ \hline\end{array}$	$\begin{array}{c c} \text{Group} & \text{Dimension} \\ \hline UT(p, q) & p^2 + q^2 + pq \\ HT(p, q) & p(p + q) \\ UT(p, q, r) & p^2 + q^2 + r^2 + pq + pr + qr \\ Sol(n) & n(n + 1)/2 \\ Nil(n) & n(n - 1)/2 \\ A(p, q) & pq \\ \hline\end{array}$
gilmore-lie-groups_ TAB_004	62	1.000	$\begin{tabular}{t}{ l l l }\hline & \text{Transformation} & \text{F} = d\mathbf{p}/dt & (\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}) A_\mu = -\frac{4\pi}{c} j_\mu \\ \hline Galilean & invariant & not invariant \\ Poincaré & not invariant & invariant \\ \hline\end{tabular}$	$\begin{array}{c cc} \text{Transformation} & \text{F} = d\mathbf{p}/dt & \left(\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}\right) A_\mu = -\frac{4\pi}{c} j_\mu \\ \hline Galilean & invariant & not invariant \\ Poincaré & not invariant & invariant \\ \hline\end{array}$	$\begin{array}{c cc} \text{Transformation} & \text{F} = d\mathbf{p}/dt & (\nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}) A_\mu = -\frac{4\pi}{c} j_\mu \\ \hline Galilean & invariant & not invariant \\ Poincaré & not invariant & invariant \\ \hline\end{array}$

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gilmore-lie-groups_ TAB_010	167	■ 0.608	$\begin{tabular}{t}{ l l l } \hline \left[H_1, H_2\right] \&= \& 0 \\ \left[\mathbf{H}, E_{\pm 2\mathbf{e}_1}\right] \&= \& (\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1} \\ \left[\mathbf{H}, E_{\pm 2\mathbf{e}_2}\right] \&= \& (0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2} \\ \left[\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}\right] \&= \& (\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2} \\ \left[E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}\right] \&= \& (2/\sqrt{12})H_1 \\ \left[E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}\right] \&= \& (2/\sqrt{12})H_2 \\ \left[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}\right] \&= \& (1/\sqrt{12})(\pm H_1 \pm H_2) \\ \left[E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}\right] \&= \& *(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2} \\ \left[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}\right] \&= \& (1/\sqrt{6})E_{-2\mathbf{e}_1} \\ \left[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}\right] \&= \& (1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ \left[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}\right] \&= \& (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \\ \left[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}\right] \&= \& (-1/\sqrt{6})E_{+2\mathbf{e}_1} \\ \left[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}\right] \&= \& (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ \left[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}\right] \&= \& *(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2} \\ \left[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}\right] \&= \& (1/\sqrt{6})E_{-2\mathbf{e}_2} \\ \left[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}\right] \&= \& (1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2} \\ \left[E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}\right] \&= \& (-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2} \\ \left[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}\right] \&= \& (-1/\sqrt{6})E_{+2\mathbf{e}_2} \\ \left[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}\right] \&= \& (-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2} \end{tabular}$	<table><tr><td>$[H_1, H_2]$</td><td>=</td><td>0</td></tr><tr><td>$[\mathbf{H}, E_{\pm 2\mathbf{e}_1}]$</td><td>=</td><td>$(\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1}$</td></tr><tr><td>$[\mathbf{H}, E_{\pm 2\mathbf{e}_2}]$</td><td>=</td><td>$(0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2}$</td></tr><tr><td>$[\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}]$</td><td>=</td><td>$(\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}$</td></tr><tr><td>$[E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}]$</td><td>=</td><td>$(2/\sqrt{12})H_1$</td></tr><tr><td>$[E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}]$</td><td>=</td><td>$(2/\sqrt{12})H_2$</td></tr><tr><td>$[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}]$</td><td>=</td><td>$(1/\sqrt{12})(\pm H_1 \pm H_2)$</td></tr><tr><td>$[E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}]$</td><td>=</td><td>$*(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$</td><td>=</td><td>$(1/\sqrt{6})E_{-2\mathbf{e}_1}$</td></tr><tr><td>$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}]$</td><td>=</td><td>$(1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{+2\mathbf{e}_1}$</td></tr><tr><td>$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$</td><td>=</td><td>$*(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2}$</td></tr><tr><td>$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}]$</td><td>=</td><td>$(1/\sqrt{6})E_{-2\mathbf{e}_2}$</td></tr><tr><td>$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}]$</td><td>=</td><td>$(1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2}$</td></tr><tr><td>$[E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$</td></tr><tr><td>$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{+2\mathbf{e}_2}$</td></tr><tr><td>$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$</td><td>=</td><td>$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$</td></tr></table>	$[H_1, H_2]$	=	0	$[\mathbf{H}, E_{\pm 2\mathbf{e}_1}]$	=	$(\pm 2/\sqrt{12}, 0)E_{\pm 2\mathbf{e}_1}$	$[\mathbf{H}, E_{\pm 2\mathbf{e}_2}]$	=	$(0, \pm 2/\sqrt{12})E_{\pm 2\mathbf{e}_2}$	$[\mathbf{H}, E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}]$	=	$(\pm/\sqrt{12}, \pm/\sqrt{12})E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}$	$[E_{+2\mathbf{e}_1}, E_{-2\mathbf{e}_1}]$	=	$(2/\sqrt{12})H_1$	$[E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(2/\sqrt{12})H_2$	$[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}]$	=	$(1/\sqrt{12})(\pm H_1 \pm H_2)$	$[E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}]$	=	$*(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(1/\sqrt{6})E_{-2\mathbf{e}_1}$	$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}]$	=	$(1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+2\mathbf{e}_1}$	$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$	=	$*(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2}$	$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(1/\sqrt{6})E_{-2\mathbf{e}_2}$	$[E_{+\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_2}]$	=	$(1/\sqrt{6})E_{+\mathbf{e}_1 + \mathbf{e}_2}$	$[E_{-2\mathbf{e}_2}, E_{+\mathbf{e}_1 + \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$	$[E_{+\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 + \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{+2\mathbf{e}_2}$	$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{-\mathbf{e}_1 - 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$[E_{+2\mathbf{e}_2}, E_{-2\mathbf{e}_2}]$	=	$(2/\sqrt{12})H_2$																																																											
$[E_{\pm \mathbf{e}_1 \pm \mathbf{e}_2}, E_{-(\pm \mathbf{e}_1 \pm \mathbf{e}_2)}]$	=	$(1/\sqrt{12})(\pm H_1 \pm H_2)$																																																											
$[E_{+2\mathbf{e}_1}, E_{-(\mathbf{e}_1 + \mathbf{e}_2)}]$	=	$*(1/\sqrt{6})E_{\mathbf{e}_1 - \mathbf{e}_2}$																																																											
$[E_{-\mathbf{e}_1 + \mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(1/\sqrt{6})E_{-2\mathbf{e}_1}$																																																											
$[E_{-\mathbf{e}_1 - \mathbf{e}_2}, E_{+2\mathbf{e}_1}]$	=	$(1/\sqrt{6})E_{+\mathbf{e}_1 - \mathbf{e}_2}$																																																											
$[E_{-2\mathbf{e}_1}, E_{\mathbf{e}_1 - \mathbf{e}_2}]$	=	$(-1/\sqrt{6})E_{-\mathbf{e}_1 - \mathbf{e}_2}$																																																											
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$[E_{+2\mathbf{e}_2}, E_{-\mathbf{e}_1 - \mathbf{e}_2}]$	=	$*(1/\sqrt{6})E_{-\mathbf{e}_1 + \mathbf{e}_2}$																																																											
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Identifier	Page	Conf.	LaTeX source	Rendered	Image
gilmore-lie-groups_TAB_011	172	■ 1.000	<pre> \begin{tabular}[t]{l l l l l l} \hline \multirow{b}{3}{*}{\((i \ \phi)^{j} \ f_{j}\)} & \multicolumn{4}{ c }{Representation} \\ \hline & \frac{1}{2} & 1 & \frac{3}{2} & 2 \\ & 4 & 3 & 4 & 5 \\ \hline & f_0 & \cos(\phi/2) & 1 & \frac{9}{8} \cos(\frac{\phi}{2}) - \frac{1}{8} \cos(\frac{3\phi}{2}) & 1 \\ (i \ \phi)^1 f_1 & 2i \sin(\phi/2) & i \sin(\phi) & \frac{9i}{4} \sin(\frac{\phi}{2}) - \frac{i}{12} \sin(\frac{3\phi}{2}) & \frac{i}{3} \sin(\phi) - \frac{i}{6} \sin(2\phi) \\ (i \ \phi)^2 f_2 & & \cos(\phi) - 1 & -\frac{1}{2} \cos(\frac{\phi}{2}) + \frac{1}{2} \cos(\frac{3\phi}{2}) & -\frac{5}{4} + \frac{1}{3} \cos(\phi) - \frac{1}{12} \cos(2\phi) \\ (i \ \phi)^3 f_3 & & & -i \sin(\frac{\phi}{2}) + \frac{i}{3} \sin(\frac{3\phi}{2}) & -\frac{i}{3} \sin(\phi) + \frac{i}{6} \sin(2\phi) \\ (i \ \phi)^4 f_4 & & & & \frac{1}{4} - \frac{1}{3} \cos(\phi) + \frac{1}{12} \cos(2\phi) \\ \hline \hline </pre>	(not rendered)	
gilmore-lie-groups_TAB_012	175	■ 0.995	<pre> \begin{tabular}[t]{l l l l l l} \hline \cos^2(\alpha, \beta) & \theta(\alpha, \beta) & n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha} & n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta} & \frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n} \\ \hline 1 & \frac{\pi}{2} \pm \frac{\pi}{2} & \pm 2 & \pm 2 & 1 \\ \frac{3}{4} & \frac{\pi}{2} \pm \frac{\pi}{3} & \pm 3 & \pm 1 & 3^{-1} \\ \frac{2}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 1 & \pm 3 & 3^{+1} \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{6} & \pm 1 & \pm 1 & 2^{-1} \\ 0 & \frac{\pi}{2} & 0 & 0 & 2^{+1} \\ \hline \hline </pre>	<pre> \begin{tabular}[t]{l l l l l l} \hline \cos^2(\alpha, \beta) & \theta(\alpha, \beta) & n = \frac{2\alpha \cdot \beta}{\alpha \cdot \alpha} & n' = \frac{2\alpha \cdot \beta}{\beta \cdot \beta} & \frac{\alpha \cdot \alpha}{\beta \cdot \beta} = \frac{n'}{n} \\ \hline 1 & \frac{\pi}{2} \pm \frac{\pi}{2} & \pm 2 & \pm 2 & 1 \\ 3 & \frac{\pi}{2} \pm \frac{\pi}{3} & \pm 3 & \pm 1 & 3^{-1} \\ \frac{2}{4} & \frac{\pi}{2} \pm \frac{\pi}{4} & \pm 1 & \pm 1 & 3^{+1} \\ \frac{1}{4} & \frac{\pi}{2} \pm \frac{\pi}{6} & \pm 1 & \pm 1 & 2^{-1} \\ 0 & \frac{\pi}{2} & 0 & 0 & 2^{+1} \\ \hline \hline </pre>	

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gilmore-lie-groups_ TAB_013	180	0.508	$\begin{tabular}{t}{ l l l l l l }\hline & \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_{l-1} & \alpha_l \\ \hline A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & " & " & " & \cdots & " & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & " & " & " & \cdots & " & \mathbf{e}_l \\ C_l & " & " & " & \cdots & " & 2\mathbf{e}_l \\ \hline \end{tabular}$	<table><tr><td></td><td>α_1</td><td>α_2</td><td>α_3</td><td></td><td>α_{l-1}</td><td>α_l</td></tr><tr><td>A_{l-1}</td><td>$\mathbf{e}_1 - \mathbf{e}_2$</td><td>$\mathbf{e}_2 - \mathbf{e}_3$</td><td>$\mathbf{e}_3 - \mathbf{e}_4$</td><td>...</td><td>$\mathbf{e}_{l-1} - \mathbf{e}_l$</td><td></td></tr><tr><td>D_l</td><td>"</td><td>"</td><td>"</td><td>...</td><td>"</td><td>$\mathbf{e}_{l-1} + \mathbf{e}_l$</td></tr><tr><td>$B_l$</td><td>"</td><td>"</td><td>"</td><td>...</td><td>"</td><td>$\mathbf{e}_l$</td></tr><tr><td>C_l</td><td>"</td><td>"</td><td>"</td><td>...</td><td>"</td><td>$2\mathbf{e}_l$</td></tr></table>		α_1	α_2	α_3		α_{l-1}	α_l	A_{l-1}	$\mathbf{e}_1 - \mathbf{e}_2$	$\mathbf{e}_2 - \mathbf{e}_3$	$\mathbf{e}_3 - \mathbf{e}_4$	...	$\mathbf{e}_{l-1} - \mathbf{e}_l$		D_l	"	"	"	...	"	$\mathbf{e}_{l-1} + \mathbf{e}_l$	B_l	"	"	"	...	"	$\mathbf{e}_l$	C_l	"	"	"	...	"	$2\mathbf{e}_l$	$\begin{array}{ccccccc} & \alpha_1 & \alpha_2 & \alpha_3 & \cdots & \alpha_{l-1} & \alpha_l \\ A_{l-1} & \mathbf{e}_1 - \mathbf{e}_2 & \mathbf{e}_2 - \mathbf{e}_3 & \mathbf{e}_3 - \mathbf{e}_4 & \cdots & \mathbf{e}_{l-1} - \mathbf{e}_l & \\ D_l & " & " & " & \cdots & " & \mathbf{e}_{l-1} + \mathbf{e}_l \\ B_l & " & " & " & \cdots & " & \mathbf{e}_l \\ C_l & " & " & " & \cdots & " & 2\mathbf{e}_l \end{array}$					
	α_1	α_2	α_3		α_{l-1}	α_l																																							
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C_l	"	"	"	...	"	$2\mathbf{e}_l$																																							
gilmore-lie-groups_ TAB_014	182	0.999	$\begin{tabular}{t}{ l l l l }\hline & (p) & (q) & (r) \\ \hline & \text{Root space} & & \\ \hline 3 & 3 & 2 & E_6 \\ 4 & 3 & 2 & E_7 \\ 5 & 3 & 2 & E_8 \\ \hline \end{tabular}$	<table><tr><td>p</td><td>q</td><td>r</td><td>Root space</td></tr><tr><td>p</td><td>2</td><td>2</td><td>D_{p+2}</td></tr><tr><td>3</td><td>3</td><td>2</td><td>E_6</td></tr><tr><td>4</td><td>3</td><td>2</td><td>E_7</td></tr><tr><td>5</td><td>3</td><td>2</td><td>E_8</td></tr></table>	p	q	r	Root space	p	2	2	D_{p+2}	3	3	2	E_6	4	3	2	E_7	5	3	2	E_8	<table><tr><th>p</th><th>q</th><th>r</th><th>Root space</th></tr><tr><td>p</td><td>2</td><td>2</td><td>D_{p+2}</td></tr><tr><td>3</td><td>3</td><td>2</td><td>E_6</td></tr><tr><td>4</td><td>3</td><td>2</td><td>E_7</td></tr><tr><td>5</td><td>3</td><td>2</td><td>E_8</td></tr></table>	p	q	r	Root space	p	2	2	D_{p+2}	3	3	2	E_6	4	3	2	E_7	5	3	2	E_8
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gilmore-lie-groups_ TAB_015	184	1.000	$\begin{tabular}{t}{ l l l l }\hline & \text{Canonical form} & \text{Boson operators} & \text{Coordinates and derivatives} \\ \hline H_i & b_i^\dagger b_i & x^i \partial_i & f_i^\dagger f_i \\ E_{+\mathbf{e}_i - \mathbf{e}_j} & b_i^\dagger b_j & x^i \partial_j & f_i^\dagger f_j \\ \hline \end{tabular}$	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i$</td></tr><tr><td>$E_{+\mathbf{e}_i - \mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	$b_i^\dagger b_i$	$x^i \partial_i$	$f_i^\dagger f_i$	$E_{+\mathbf{e}_i - \mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	<table><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i$</td></tr><tr><td>$E_{+\mathbf{e}_i - \mathbf{e}_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td></tr></table>	Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	H_i	$b_i^\dagger b_i$	$x^i \partial_i$	$f_i^\dagger f_i$	$E_{+\mathbf{e}_i - \mathbf{e}_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$																
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Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																
gilmore-lie-groups_ TAB_016	184	0.960	<pre>\begin{tabular}[t]{ l l l l l } \hline \multirow[b]{2}{*}{Canonical form} & \multicolumn{2}{ c }{\((C_l)\)} & \multicolumn{2}{ c }{\((D_l)\)} \\ \hline & \text{Boson operators} & \text{Coordinates and derivatives} & \text{Fermion operators} & \text{Coordinates and derivatives} \\ \hline H_i & b_i^\dagger b_i + \frac{1}{2} & x^i \partial_i & f_i^\dagger f_i + \frac{1}{2} & x^i \partial_i + \frac{1}{2} \\ E_{+e_i - e_j} & b_i^\dagger b_j & x^i \partial_j & f_i^\dagger f_j & x^i \partial_j \\ E_{+e_i + e_j} & b_i^\dagger b_j^\dagger & x^i x^j & f_i^\dagger f_j^\dagger & x^i x^j \\ E_{-e_i - e_j} & b_i b_j & \partial_i \partial_j & f_i f_j & \partial_i \partial_j \\ E_{+2e_i} & b_i^\dagger b_i^\dagger & x^i x^i & & \\ E_{-2e_i} & b_i b_i & \partial_i \partial_i & & \end{tabular}</pre>	(not rendered)	<table><tr><th colspan="2"></th><th colspan="2">C_l</th><th colspan="2">D_l</th></tr><tr><th>Canonical form</th><th>Boson operators</th><th>Coordinates and derivatives</th><th>Fermion operators</th><th>Coordinates and derivatives</th><th></th></tr><tr><td>H_i</td><td>$b_i^\dagger b_i + \frac{1}{2}$</td><td>$x^i \partial_i$</td><td>$f_i^\dagger f_i + \frac{1}{2}$</td><td>$x^i \partial_i + \frac{1}{2}$</td><td></td></tr><tr><td>$E_{+e_i - e_j}$</td><td>$b_i^\dagger b_j$</td><td>$x^i \partial_j$</td><td>$f_i^\dagger f_j$</td><td>$x^i \partial_j$</td><td></td></tr><tr><td>$E_{+e_i + e_j}$</td><td>$b_i^\dagger b_j^\dagger$</td><td>$x^i x^j$</td><td>$f_i^\dagger f_j^\dagger$</td><td>$x^i x^j$</td><td></td></tr><tr><td>$E_{-e_i - e_j}$</td><td>$b_i b_j$</td><td>$\partial_i \partial_j$</td><td>$f_i f_j$</td><td>$\partial_i \partial_j$</td><td></td></tr><tr><td>E_{+2e_i}</td><td>$b_i^\dagger b_i^\dagger$</td><td>$x^i x^i$</td><td></td><td></td><td></td></tr><tr><td>E_{-2e_i}</td><td>$b_i b_i$</td><td>$\partial_i \partial_i$</td><td></td><td></td><td></td></tr></table>			C_l		D_l		Canonical form	Boson operators	Coordinates and derivatives	Fermion operators	Coordinates and derivatives		H_i	$b_i^\dagger b_i + \frac{1}{2}$	$x^i \partial_i$	$f_i^\dagger f_i + \frac{1}{2}$	$x^i \partial_i + \frac{1}{2}$		$E_{+e_i - e_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	$x^i \partial_j$		$E_{+e_i + e_j}$	$b_i^\dagger b_j^\dagger$	$x^i x^j$	$f_i^\dagger f_j^\dagger$	$x^i x^j$		$E_{-e_i - e_j}$	$b_i b_j$	$\partial_i \partial_j$	$f_i f_j$	$\partial_i \partial_j$		E_{+2e_i}	$b_i^\dagger b_i^\dagger$	$x^i x^i$				E_{-2e_i}	$b_i b_i$	$\partial_i \partial_i$			
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$E_{+e_i - e_j}$	$b_i^\dagger b_j$	$x^i \partial_j$	$f_i^\dagger f_j$	$x^i \partial_j$																																																	
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gilmore-lie-groups_ TAB_017	195	0.899	<pre>\begin{tabular}[t]{ l l l l l } \hline Mapping & Real form & Root space & Condition \\ \hline Block submatrix & $\mathfrak{so}(p, q)$ & D_n & $p + q = 2n$ \\ & $\mathfrak{so}(p, q)$ & B_n & $p + q = 2n + 1$ \\ & $\mathfrak{su}(p, q)$ & A_{n-1} & $p + q = n$ \\ & $\mathfrak{sp}(p, q)$ & C_n & $p + q = n$ \\ Subfield restriction & $\mathfrak{sl}(n; \mathbb{R})$ & A_{n-1} & \\ & $\mathfrak{sp}(2n; \mathbb{R})$ & C_n & \\ Field embedding & $\mathfrak{so}^*(2n)$ & D_n & \\ & $\mathfrak{su}^*(2n)$ & A_{2n-1} & \end{tabular}</pre>	(not rendered)	<table><tr><th>Mapping</th><th>Real form</th><th>Root space</th><th>Condition</th></tr><tr><td>Block submatrix</td><td>$\mathfrak{so}(p, q)$</td><td>D_n</td><td>$p + q = 2n$</td></tr><tr><td></td><td>$\mathfrak{so}(p, q)$</td><td>B_n</td><td>$p + q = 2n + 1$</td></tr><tr><td></td><td>$\mathfrak{su}(p, q)$</td><td>A_{n-1}</td><td>$p + q = n$</td></tr><tr><td></td><td>$\mathfrak{sp}(p, q)$</td><td>C_n</td><td>$p + q = n$</td></tr><tr><td>Subfield restriction</td><td>$\mathfrak{sl}(n; \mathbb{R})$</td><td>$A_{n-1}$</td><td></td></tr><tr><td></td><td>$\mathfrak{sp}(2n; \mathbb{R})$</td><td>$C_n$</td><td></td></tr><tr><td>Field embedding</td><td>$\mathfrak{so}^*(2n)$</td><td>D_n</td><td></td></tr><tr><td></td><td>$\mathfrak{su}^*(2n)$</td><td>A_{2n-1}</td><td></td></tr></table>	Mapping	Real form	Root space	Condition	Block submatrix	$\mathfrak{so}(p, q)$	D_n	$p + q = 2n$		$\mathfrak{so}(p, q)$	B_n	$p + q = 2n + 1$		$\mathfrak{su}(p, q)$	A_{n-1}	$p + q = n$		$\mathfrak{sp}(p, q)$	C_n	$p + q = n$	Subfield restriction	$\mathfrak{sl}(n; \mathbb{R})$	A_{n-1}			$\mathfrak{sp}(2n; \mathbb{R})$	C_n		Field embedding	$\mathfrak{so}^*(2n)$	D_n			$\mathfrak{su}^*(2n)$	A_{2n-1}													
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gilmore-lie-groups_ TAB_018	195	■ 0.500	$\begin{tabular}{t l l l l l } \hline \scriptstyle (A_1) & \sim & \scriptstyle (B_1) & \sim & \scriptstyle (C_1) \\ \scriptstyle (B_1) & \sim & \scriptstyle (C_1) & \sim & \scriptstyle (C_1) \\ \hline \scriptstyle (u(2)) & \sim & \scriptstyle (so(3)) & \sim & \scriptstyle (sp(1) = usp(2)) \\ \scriptstyle (u(1,1) = sl(2; \mathbb{R})) & \sim & \scriptstyle (so(2,1)) & \sim & \scriptstyle (sp(2; \mathbb{R})) \\ \hline \scriptstyle (D_2) & = & \scriptstyle (A_1) & + & \scriptstyle (A_1) \\ \scriptstyle (so(4)) & = & \scriptstyle (so(3)) & + & \scriptstyle (so(3)) \\ \scriptstyle (so^*(4)) & \sim & \scriptstyle (so(3)) & + & \scriptstyle (so(2,1)) \\ \scriptstyle (so(3,1)) & \sim & \scriptstyle (sl(2; \mathbb{C})) & & \\ \scriptstyle (so(2,2)) & \sim & \scriptstyle (so(2,1)) & + & \scriptstyle (so(2,1)) \\ \hline \scriptstyle (B_2) & = & \scriptstyle (C_2) & & \\ \scriptstyle (so(5)) & \sim & \scriptstyle (sp(2) = usp(4)) & & \\ \scriptstyle (so(4,1)) & \sim & \scriptstyle (sp(1,1) = usp(2,2)) & & \\ \scriptstyle (so(3,2)) & \sim & \scriptstyle (sp(4; R)) & & \\ \hline \scriptstyle (D_3) & = & \scriptstyle (A_3) & & \\ \scriptstyle (so(6)) & \sim & \scriptstyle (su(4)) & & \\ \scriptstyle (so(5,1)) & \sim & \scriptstyle (su^*(4)) & & \\ \scriptstyle (so^*(6)) & \sim & \scriptstyle (su(3,1)) & & \\ \scriptstyle (so(4,2)) & \sim & \scriptstyle (su(2,2)) & & \\ \scriptstyle (so(3,3)) & \sim & \scriptstyle (sl(4; \mathbb{R})) & & \\ \hline \end{tabular}$	<table><tr><td>A_1</td><td></td><td>B_1</td><td></td><td>C_1</td></tr><tr><td>$su(2)$</td><td></td><td>$so(3)$</td><td></td><td>$sp(1) = usp(2)$</td></tr><tr><td>$su(1, 1) = sl(2; \mathbb{R})$</td><td></td><td>$so(2, 1)$</td><td></td><td>$sp(2; \mathbb{R})$</td></tr><tr><td>$D_2$</td><td>$=$</td><td>$A_1$</td><td>$+$</td><td>$A_1$</td></tr><tr><td>$so(4)$</td><td>$=$</td><td>$so(3)$</td><td>$+$</td><td>$so(3)$</td></tr><tr><td>$so^*(4)$</td><td></td><td>$so(3)$</td><td>$+$</td><td>$so(2, 1)$</td></tr><tr><td>$so(3, 1)$</td><td></td><td>$sl(2; \mathbb{C})$</td><td></td><td></td></tr><tr><td>$so(2, 2)$</td><td></td><td>$so(2, 1)$</td><td>$+$</td><td>$so(2, 1)$</td></tr><tr><td>B_2</td><td>$=$</td><td>C_2</td><td></td><td></td></tr><tr><td>$so(5)$</td><td>$\sim$</td><td>$sp(2) = usp(4)$</td><td></td><td></td></tr><tr><td>$so(4, 1)$</td><td>$\sim$</td><td>$sp(1, 1) = usp(2, 2)$</td><td></td><td></td></tr><tr><td>$so(3, 2)$</td><td>$\sim$</td><td>$sp(4; R)$</td><td></td><td></td></tr><tr><td>D_3</td><td>$=$</td><td>A_3</td><td></td><td></td></tr><tr><td>$so(6)$</td><td>$\sim$</td><td>$su(4)$</td><td></td><td></td></tr><tr><td>$so(5, 1)$</td><td>$\sim$</td><td>$su^*(4)$</td><td></td><td></td></tr><tr><td>$so^*(6)$</td><td>$\sim$</td><td>$su(3, 1)$</td><td></td><td></td></tr><tr><td>$so(4, 2)$</td><td>$\sim$</td><td>$su(2, 2)$</td><td></td><td></td></tr><tr><td>$so(3, 3)$</td><td>$\sim$</td><td>$sl(4; \mathbb{R})$</td><td></td><td></td></tr></table>	A_1		B_1		C_1	$su(2)$		$so(3)$		$sp(1) = usp(2)$	$su(1, 1) = sl(2; \mathbb{R})$		$so(2, 1)$		$sp(2; \mathbb{R})$	D_2	$=$	A_1	$+$	A_1	$so(4)$	$=$	$so(3)$	$+$	$so(3)$	$so^*(4)$		$so(3)$	$+$	$so(2, 1)$	$so(3, 1)$		$sl(2; \mathbb{C})$			$so(2, 2)$		$so(2, 1)$	$+$	$so(2, 1)$	B_2	$=$	C_2			$so(5)$	$\sim$	$sp(2) = usp(4)$			$so(4, 1)$	$\sim$	$sp(1, 1) = usp(2, 2)$			$so(3, 2)$	$\sim$	$sp(4; R)$			D_3	$=$	A_3			$so(6)$	$\sim$	$su(4)$			$so(5, 1)$	$\sim$	$su^*(4)$			$so^*(6)$	$\sim$	$su(3, 1)$			$so(4, 2)$	$\sim$	$su(2, 2)$			$so(3, 3)$	$\sim$	$sl(4; \mathbb{R})$			
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$so(4, 1)$	$\sim$	$sp(1, 1) = usp(2, 2)$																																																																																													
$so(3, 2)$	$\sim$	$sp(4; R)$																																																																																													
D_3	$=$	A_3																																																																																													
$so(6)$	$\sim$	$su(4)$																																																																																													
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$so(4, 2)$	$\sim$	$su(2, 2)$																																																																																													
$so(3, 3)$	$\sim$	$sl(4; \mathbb{R})$																																																																																													

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gilmore-lie-groups_ TAB_019	196	■1.000	<pre>\begin{tabular}[t]{ l l l l } \hline \multirow[b]{2}{*}{Root space} & \multirow[b]{2}{*}{Class \\ ({ }_ {\text {rank(character) }})} & \multicolumn{2}{ l }{Maximal compact subgroup} \\ \hline & & Root space & Dimension \\ \hline G_2 & G_{2(-14)} & G_2 & 14 \\ & G_{2(+2)} & A_1 + A_1 & 6 \\ F_4 & F_{4(-52)} & F_4 & 52 \\ & F_{4(-20)} & B_4 & 36 \\ & F_{4(+4)} & C_3 + A_1 & 24 \\ E_6 & E_{6(-78)} & E_6 & 78 \\ & E_{6(-26)} & F_4 & 52 \\ & E_{6(-14)} & D_5 + D_1 & 46 \\ & E_{6(+2)} & A_5 + A_1 & 38 \\ & E_{6(+6)} & C_4 & 36 \\ E_7 & E_{7(-133)} & E_7 & 133 \\ & E_{7(-25)} & E_6 + D_1 & 79 \\ & E_{7(-5)} & D_6 + A_1 & 69 \\ & E_{7(+7)} & A_7 & 63 \\ E_8 & E_{8(-248)} & E_8 & 248 \\ & E_{8(-24)} & E_7 + A_1 & 136 \\ & E_{8(+8)} & D_8 & 120 \\ \hline \hline \end{tabular}</pre>	(not rendered)	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$															
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III	J_-	$n_2 = -1$																																	
gilmore-lie-groups_ TAB_020	199	■1.000	<pre>\begin{tabular}[t]{ l l l l } \hline Quadrant & Operator & Edge \\ \hline I & \((J_{+})\) & \((n_{2}=0)\) \\ \hline I & \((J_{-})\) & \((n_{1}=0)\) \\ \hline III & \((J_{+})\) & \((n_{1}=-1)\) \\ \hline III & \((J_{-})\) & \((n_{2}=-1)\) \\ \hline \end{tabular}</pre>	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$	<table><tr><th>Quadrant</th><th>Operator</th><th>Edge</th></tr><tr><td>I</td><td>J_+</td><td>$n_2 = 0$</td></tr><tr><td>I</td><td>J_-</td><td>$n_1 = 0$</td></tr><tr><td>III</td><td>J_+</td><td>$n_1 = -1$</td></tr><tr><td>III</td><td>J_-</td><td>$n_2 = -1$</td></tr></table>	Quadrant	Operator	Edge	I	J_+	$n_2 = 0$	I	J_-	$n_1 = 0$	III	J_+	$n_1 = -1$	III	J_-	$n_2 = -1$
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Identifier	Page	Conf.	LaTeX source	Rendered	Image																																																																														
gilmore-lie-groups_021	206	0.877	$\begin{tabular}{t l l l l l } \hline \text{Root space} \& \text{Quotient } (G^{\prime}/H) \& \text{Dimension } (P) \& \text{Rank } (P) \\ \hline (A_{p+q-1}) \& (SU(p, q)/S[U(p) \otimes U(q)]) \& (2pq) \& \min(p, q) \\ (A_{n-1}) \& (SL(n; \mathbb{R})/SO(n)) \& (\frac{1}{2}(n+2)(n-1)) \& n-1 \\ (A_{2n-1}) \& (SU^*(2n)/USp(2n)) \& ((2n+1)(n-1)) \& n-1 \\ (B_{p+q}) \& (SO(p, q)/SO(p) \otimes SO(q)) \& (pq) \& \min(p, q) \\ (D_{p+q}) \& (SO(p, q)/SO(p) \otimes SO(q)) \& (pq) \& \min(p, q) \\ (D_n) \& (SO^*(2n)/U(n)) \& (n(n-1)) \& [n/2] \\ (C_{p+q}) \& (USp(2p, 2q)/USp(2p) \otimes USp(2q)) \& (4pq) \& \min(p, q) \\ (C_n) \& (Sp(2n; \mathbb{R})/U(n)) \& (n(n+1)) \& n \\ \hline \end{tabular}$	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p, q)/S[U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p, q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n; \mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p, 2q)/USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p, q)/S[U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$	A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n	<table><tr><th>Root space</th><th>Quotient G'/H</th><th>Dimension P</th><th>Rank P</th></tr><tr><td>A_{p+q-1}</td><td>$SU(p, q)/S[U(p) \otimes U(q)]$</td><td>$2pq$</td><td>$\min(p, q)$</td></tr><tr><td>$A_{n-1}$</td><td>$SL(n; \mathbb{R})/SO(n)$</td><td>$\frac{1}{2}(n+2)(n-1)$</td><td>$n-1$</td></tr><tr><td>$A_{2n-1}$</td><td>$SU^*(2n)/USp(2n)$</td><td>$(2n+1)(n-1)$</td><td>$n-1$</td></tr><tr><td>$B_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_{p+q}$</td><td>$SO(p, q)/SO(p) \otimes SO(q)$</td><td>$pq$</td><td>$\min(p, q)$</td></tr><tr><td>$D_n$</td><td>$SO^*(2n)/U(n)$</td><td>$n(n-1)$</td><td>$[n/2]$</td></tr><tr><td>$C_{p+q}$</td><td>$USp(2p, 2q)/USp(2p) \otimes USp(2q)$</td><td>$4pq$</td><td>$\min(p, q)$</td></tr><tr><td>$C_n$</td><td>$Sp(2n; \mathbb{R})/U(n)$</td><td>$n(n+1)$</td><td>$n$</td></tr></table>	Root space	Quotient G'/H	Dimension P	Rank P	A_{p+q-1}	$SU(p, q)/S[U(p) \otimes U(q)]$	$2pq$	$\min(p, q)$	A_{n-1}	$SL(n; \mathbb{R})/SO(n)$	$\frac{1}{2}(n+2)(n-1)$	$n-1$	A_{2n-1}	$SU^*(2n)/USp(2n)$	$(2n+1)(n-1)$	$n-1$	B_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_{p+q}	$SO(p, q)/SO(p) \otimes SO(q)$	pq	$\min(p, q)$	D_n	$SO^*(2n)/U(n)$	$n(n-1)$	$[n/2]$	C_{p+q}	$USp(2p, 2q)/USp(2p) \otimes USp(2q)$	$4pq$	$\min(p, q)$	C_n	$Sp(2n; \mathbb{R})/U(n)$	$n(n+1)$	n						
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gilmore-lie-groups_022	207	0.979	$\begin{tabular}{t l l l l l l } \hline \text{Root space} \& (G^{\prime}/H) \& (\operatorname{Dim} G^{\prime}) \& (\operatorname{Dim} H) \& (\operatorname{Dim} P) \& \text{Rank } (P) \\ \hline (G_2) \& (G_{2(+2)}) \& 14 \& 6 \& 8 \& 2 \\ (F_4) \& (F_{4(-20)}) \& 52 \& 36 \& 16 \& 1 \\ (F_4) \& (F_{4(+4)}) \& 52 \& 24 \& 28 \& 4 \\ (E_6) \& (E_{6(-26)}) \& 78 \& 52 \& 26 \& 2 \\ (E_6) \& (E_{6(-14)}) \& 78 \& 46 \& 32 \& 2 \\ (E_6) \& (E_{6(+2)}) \& 78 \& 38 \& 40 \& 4 \\ (E_6) \& (E_{6(+6)}) \& 78 \& 36 \& 42 \& 6 \\ (E_7) \& (E_{7(-25)}) \& 133 \& 79 \& 54 \& 3 \\ (E_7) \& (E_{7(-5)}) \& 133 \& 69 \& 64 \& 4 \\ (E_7) \& (E_{7(+7)}) \& 133 \& 63 \& 70 \& 7 \\ (E_8) \& (E_{8(-24)}) \& 248 \& 136 \& 112 \& 4 \\ (E_8) \& (E_{8(+8)}) \& 248 \& 120 \& 128 \& 8 \\ \hline \end{tabular}$	(not rendered)	<table><tr><th>Root space</th><th>G'/H</th><th>Dim G'</th><th>Dim H</th><th>Dim P</th><th>Rank P</th></tr><tr><td>G_2</td><td>$G_{2(+2)}/(A_1 \oplus A_1)$</td><td>14</td><td>6</td><td>8</td><td>2</td></tr><tr><td>F_4</td><td>$F_{4(-20)}/B_4$</td><td>52</td><td>36</td><td>16</td><td>1</td></tr><tr><td></td><td>$F_{4(+4)}/(C_3 \oplus A_1)$</td><td>52</td><td>24</td><td>28</td><td>4</td></tr><tr><td>E_6</td><td>$E_{6(-26)}/F_4$</td><td>78</td><td>52</td><td>26</td><td>2</td></tr><tr><td></td><td>$E_{6(-14)}/(D_5 \oplus D_1)$</td><td>78</td><td>46</td><td>32</td><td>2</td></tr><tr><td></td><td>$E_{6(+2)}/(A_5 \oplus A_1)$</td><td>78</td><td>38</td><td>40</td><td>4</td></tr><tr><td></td><td>$E_{6(+6)}/C_4$</td><td>78</td><td>36</td><td>42</td><td>6</td></tr><tr><td>E_7</td><td>$E_{7(-25)}/(E_6 \oplus D_1)$</td><td>133</td><td>79</td><td>54</td><td>3</td></tr><tr><td></td><td>$E_{7(-5)}/(D_6 \oplus A_1)$</td><td>133</td><td>69</td><td>64</td><td>4</td></tr><tr><td></td><td>$E_{7(+7)}/A_7$</td><td>133</td><td>63</td><td>70</td><td>7</td></tr><tr><td>E_8</td><td>$E_{8(-24)}/(E_7 \oplus A_1)$</td><td>248</td><td>136</td><td>112</td><td>4</td></tr><tr><td></td><td>$E_{8(+8)}/D_8$</td><td>248</td><td>120</td><td>128</td><td>8</td></tr></table>	Root space	G'/H	Dim G'	Dim H	Dim P	Rank P	G_2	$G_{2(+2)}/(A_1 \oplus A_1)$	14	6	8	2	F_4	$F_{4(-20)}/B_4$	52	36	16	1		$F_{4(+4)}/(C_3 \oplus A_1)$	52	24	28	4	E_6	$E_{6(-26)}/F_4$	78	52	26	2		$E_{6(-14)}/(D_5 \oplus D_1)$	78	46	32	2		$E_{6(+2)}/(A_5 \oplus A_1)$	78	38	40	4		$E_{6(+6)}/C_4$	78	36	42	6	E_7	$E_{7(-25)}/(E_6 \oplus D_1)$	133	79	54	3		$E_{7(-5)}/(D_6 \oplus A_1)$	133	69	64	4		$E_{7(+7)}/A_7$	133	63	70	7	E_8	$E_{8(-24)}/(E_7 \oplus A_1)$	248	136	112	4		$E_{8(+8)}/D_8$	248	120	128	8
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Identifier	Page	Conf.	LaTeX source	Rendered	Image																																
gilmore-lie-groups_ TAB_023	212	■ 1.000	$\begin{tabular}{t}{ l l l l }\hline \text{Operation} \& \backslash(i \sigma_1 \backslash) \& \backslash(i \sigma_2 \backslash) \& \backslash(i \sigma_3 \backslash) \\\hline \backslash(T_1=\backslash) \text{block matrix decomposition} \& -1 \& -1 \& +1 \\\hline \backslash(T_2=\backslash) \text{complex conjugation} \& -1 \& +1 \& -1 \\\hline \backslash(T_3=T_1 T_2) \& +1 \& -1 \& -1 \\\hline \end{tabular}$	<table><tr><th>Operation</th><th>$i\sigma_1$</th><th>$i\sigma_2$</th><th>$i\sigma_3$</th></tr><tr><td>$T_1 = \text{block matrix decomposition}$</td><td>-1</td><td>-1</td><td>+1</td></tr><tr><td>$T_2 = \text{complex conjugation}$</td><td>-1</td><td>+1</td><td>-1</td></tr><tr><td>$T_3 = T_1 T_2$</td><td>+1</td><td>-1</td><td>-1</td></tr></table>	Operation	$i\sigma_1$	$i\sigma_2$	$i\sigma_3$	$T_1 = \text{block matrix decomposition}$	-1	-1	+1	$T_2 = \text{complex conjugation}$	-1	+1	-1	$T_3 = T_1 T_2$	+1	-1	-1	<table><tr><th>Operation</th><th>$i\sigma_1$</th><th>$i\sigma_2$</th><th>$i\sigma_3$</th></tr><tr><td>$T_1 = \text{block matrix decomposition}$</td><td>-1</td><td>-1</td><td>+1</td></tr><tr><td>$T_2 = \text{complex conjugation}$</td><td>-1</td><td>+1</td><td>-1</td></tr><tr><td>$T_3 = T_1 T_2$</td><td>+1</td><td>-1</td><td>-1</td></tr></table>	Operation	$i\sigma_1$	$i\sigma_2$	$i\sigma_3$	$T_1 = \text{block matrix decomposition}$	-1	-1	+1	$T_2 = \text{complex conjugation}$	-1	+1	-1	$T_3 = T_1 T_2$	+1	-1	-1
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gilmore-lie-groups_ TAB_024	232	■ 1.000	$\begin{tabular}{t}{ l l l l }\hline \text{Bessel function} \& \rightarrow \& \text{harmonic oscillator eigenfunction} \\\hline \text{Bessel equation} \& \rightarrow \& \text{Schrödinger equation for harmonic oscilator} \\\hline \end{tabular}$	<table><tr><td>Bessel function</td><td>→</td><td>harmonic oscillator eigenfunction</td></tr><tr><td>Bessel equation</td><td>→</td><td>Schrödinger equation for harmonic oscilator</td></tr></table>	Bessel function	→	harmonic oscillator eigenfunction	Bessel equation	→	Schrödinger equation for harmonic oscilator	<table><tr><td>Bessel function</td><td>→</td><td>harmonic oscillator eigenfunction</td></tr><tr><td>Bessel equation</td><td>→</td><td>Schrödinger equation for harmonic oscilator</td></tr></table>	Bessel function	→	harmonic oscillator eigenfunction	Bessel equation	→	Schrödinger equation for harmonic oscilator																				
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gilmore-lie-groups_ TAB_025	239	■ 0.424	$\begin{tabular}{t}{ l l l l l }\hline \text{Equation} \& \backslash(A \backslash) \& \backslash(B \backslash) \& \backslash(C \backslash) \\\hline \text{Klein-Gordon} \& \backslash(-l(l+1)+\backslashleft(e^2 / \hbar c \right)^2 \backslash) \& \backslash(2 E e^2 /(\hbar c) \backslash) \& \backslash(\backslashleft[E^2-\backslashleft(m c^2 \right)^2 \right] /(\hbar c)^2 \backslash) \\\hline \text{Schrödinger} \& \backslash(-l(l+1) \backslash) \& \backslash(2 m e^2 / \hbar^2 \backslash) \& \backslash(2 \mathrm{~m} \mathrm{~W} / \hbar^2 \backslash) \\\hline \end{tabular}$	<table><tr><th>Equation</th><th>A</th><th>B</th><th>C</th></tr><tr><td>Klein-Gordon</td><td>$-l(l+1)+(e^2/\hbar c)^2$</td><td>$2Ee^2/(\hbar c)^2$</td><td>$[E^2-(mc^2)^2]/(\hbar c)^2$</td></tr><tr><td>Schrödinger</td><td>$-l(l+1)$</td><td>$2me^2/\hbar^2$</td><td>$2\mathrm{~m}\mathrm{~W}/\hbar^2$</td></tr></table>	Equation	A	B	C	Klein-Gordon	$-l(l+1)+(e^2/\hbar c)^2$	$2Ee^2/(\hbar c)^2$	$[E^2-(mc^2)^2]/(\hbar c)^2$	Schrödinger	$-l(l+1)$	$2me^2/\hbar^2$	$2\mathrm{~m}\mathrm{~W}/\hbar^2$	<table><tr><th>Equation</th><th>A</th><th>B</th><th>C</th></tr><tr><td>Klein–Gordon</td><td>$-l(l+1)+(e^2/\hbar c)^2$</td><td>$2Ee^2/(\hbar c)^2$</td><td>$[E^2-(mc^2)^2]/(\hbar c)^2$</td></tr><tr><td>Schrödinger</td><td>$-l(l+1)$</td><td>$2me^2/\hbar^2$</td><td>$2mW/\hbar^2$</td></tr></table>	Equation	A	B	C	Klein–Gordon	$-l(l+1)+(e^2/\hbar c)^2$	$2Ee^2/(\hbar c)^2$	$[E^2-(mc^2)^2]/(\hbar c)^2$	Schrödinger	$-l(l+1)$	$2me^2/\hbar^2$	$2mW/\hbar^2$								
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Identifier	Page	Conf.	LaTeX source	Rendered	Image																				
gilmore-lie-groups_ TAB_026	240	■ 0.655	<pre>\begin{tabular}[t]{ l l l } \hline & Procedure & Result \\ \hline 1 & Locate singularities & $0, \infty$ \\ 2 & Determine analytic behavior at singular points & $r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0$ \\ 3 & Keep only $\mathcal{L}^2$ solutions & $\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$ \\ 4 & Look for solutions with proper asymptotic behavior & $R = r^\gamma e^{\lambda r} f(r)$ \\ 5 & Construct differential equation for $f(r)$ & $[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)] f(r) = 0$ \\ 6 & Construct recursion relation & $f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$ \\ 7 & Look at asymptotic behavior & $f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$) \\ 8 & Construct quantization condition & $2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$ \\ 9 & Construct explicit solutions & $E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$ \\ \hline \end{tabular}</pre>	(not rendered)	<table><tr><th>Procedure</th><th>Result</th></tr><tr><td>1 Locate singularities</td><td>$0, \infty$</td></tr><tr><td>2 Determine analytic behavior at singular points</td><td>$r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0$</td></tr><tr><td>3 Keep only $\mathcal{L}^2$ solutions</td><td>$\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$</td></tr><tr><td>4 Look for solutions with proper asymptotic behavior</td><td>$R = r^\gamma e^{\lambda r} f(r)$</td></tr><tr><td>5 Construct differential equation for $f(r)$</td><td>$[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)] f(r) = 0$</td></tr><tr><td>6 Construct recursion relation</td><td>$f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$</td></tr><tr><td>7 Look at asymptotic behavior</td><td>$f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$)</td></tr><tr><td>8 Construct quantization condition</td><td>$2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$</td></tr><tr><td>9 Construct explicit solutions</td><td>$E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$</td></tr></table>	Procedure	Result	1 Locate singularities	$0, \infty$	2 Determine analytic behavior at singular points	$r \rightarrow 0 : R \simeq r^\gamma, \gamma(\gamma - 1) + A = 0$ $r \rightarrow \infty : R \simeq e^{\lambda r}, \lambda^2 + C = 0$	3 Keep only $\mathcal{L}^2$ solutions	$\gamma = \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A}, \lambda = -\sqrt{-C}$	4 Look for solutions with proper asymptotic behavior	$R = r^\gamma e^{\lambda r} f(r)$	5 Construct differential equation for $f(r)$	$[(rD^2 + 2\gamma D) + (2\lambda\gamma + B + 2\lambda r D)] f(r) = 0$	6 Construct recursion relation	$f_{j+1} = -\frac{2\lambda(j + \gamma) + B}{j(j + 1) + 2\gamma(j + 1)} f_j$	7 Look at asymptotic behavior	$f \simeq e^{-2\lambda r}$ if series does not terminate $\simeq e^{+1\lambda r}$ if series does terminate ($\lambda < 0$)	8 Construct quantization condition	$2\lambda(n + \gamma) + B = 0$ or $n + \frac{1}{2} + \sqrt{(\frac{1}{2})^2 - A} = \frac{B}{2\sqrt{-C}}$	9 Construct explicit solutions	$E = \frac{mc^2}{\sqrt{1 + (\alpha/N')^2}}, W = -\frac{1}{2}mc^2\alpha^2 \frac{1}{N^2}$ $N' = n + \frac{1}{2} + \sqrt{(l + \frac{1}{2})^2 - \alpha^2}, N = n + l + 1$
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gilmore-lie-groups_TAB_027	244	■0.998	$\begin{tabular}{t l l l l } \hline \text{\(m\)} & \text{\(l=0\)} & \text{\(l=1\)} & \text{\(l=2\)} & \text{\(l=3\)} \\ \hline 0 & \sqrt{\frac{1}{4\pi}} & \sqrt{\frac{3}{4\pi}} \cos \theta & \sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1) & \sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta) \\ \pm 1 & & \mp \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{\pm i \phi} & \mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta \, e^{\pm i \phi} & \mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) \, e^{\pm i \phi} \\ \pm 2 & & & \sqrt{\frac{15}{32\pi}} \sin^2 \theta \, e^{\pm 2i \phi} & \sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta \, e^{\pm 2i \phi} \\ \pm 3 & & & & \mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta \, e^{\pm 3i \phi} \\ \hline \hline \end{tabular}$	<table><tr><th>m</th><th>$l = 0$</th><th>$l = 1$</th><th>$l = 2$</th><th>$l = 3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}} \cos \theta$</td><td>$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$</td><td>$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) e^{\pm i \phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i \phi}$</td><td>$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2i \phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3i \phi}$</td></tr></table>	m	$l = 0$	$l = 1$	$l = 2$	$l = 3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}} \cos \theta$	$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$	$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$	± 1		$\mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta e^{\pm i \phi}$	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) e^{\pm i \phi}$	± 2			$\sqrt{\frac{15}{32\pi}} \sin^2 \theta e^{\pm 2i \phi}$	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta e^{\pm 2i \phi}$	± 3				$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta e^{\pm 3i \phi}$	<table><tr><th>m</th><th>$l = 0$</th><th>$l = 1$</th><th>$l = 2$</th><th>$l = 3$</th></tr><tr><td>0</td><td>$\sqrt{\frac{1}{4\pi}}$</td><td>$\sqrt{\frac{3}{4\pi}} \cos \theta$</td><td>$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$</td><td>$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$</td></tr><tr><td>$\pm 1$</td><td></td><td>$\mp \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta \, e^{\pm i \phi}$</td><td>$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) \, e^{\pm i \phi}$</td></tr><tr><td>$\pm 2$</td><td></td><td></td><td>$\sqrt{\frac{15}{32\pi}} \sin^2 \theta \, e^{\pm 2i \phi}$</td><td>$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta \, e^{\pm 2i \phi}$</td></tr><tr><td>$\pm 3$</td><td></td><td></td><td></td><td>$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta \, e^{\pm 3i \phi}$</td></tr></table>	m	$l = 0$	$l = 1$	$l = 2$	$l = 3$	0	$\sqrt{\frac{1}{4\pi}}$	$\sqrt{\frac{3}{4\pi}} \cos \theta$	$\sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1)$	$\sqrt{\frac{7}{16\pi}} (5 \cos^3 \theta - 3 \cos \theta)$	± 1		$\mp \sqrt{\frac{3}{8\pi}} \sin \theta \, e^{\pm i \phi}$	$\mp \sqrt{\frac{15}{8\pi}} \cos \theta \sin \theta \, e^{\pm i \phi}$	$\mp \sqrt{\frac{21}{64\pi}} \sin \theta (5 \cos^2 \theta - 1) \, e^{\pm i \phi}$	± 2			$\sqrt{\frac{15}{32\pi}} \sin^2 \theta \, e^{\pm 2i \phi}$	$\sqrt{\frac{105}{32\pi}} \sin^2 \theta \cos \theta \, e^{\pm 2i \phi}$	± 3				$\mp \sqrt{\frac{35}{64\pi}} \sin^3 \theta \, e^{\pm 3i \phi}$				
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gilmore-lie-groups_TAB_028	258	■0.721	$\begin{tabular}{t l l l } \hline H & \text{\(A_4\)} & \text{\(\Gamma_4\)} & \text{\(\Gamma_5\)} \\ \hline A & \text{\(\Theta\)} & \text{\(A\)} & \text{\(B\)} \\ \hline B & \text{\(A\)} & \text{\(B\)} & \text{\(C\)} \\ \hline C & \text{\(A\)} & \text{\(B\)} & \text{\(C\)} \\ \hline \hline \end{tabular}$	<table><tr><th>H</th><th>$\frac{p^2}{2m} - \frac{\alpha}{r}$</th><th>$\sqrt{p^2 + m^2} - \frac{\alpha}{r}$</th></tr><tr><td>$A_4$</td><td>$\mathbf{r} \cdot \mathbf{p} - i$</td><td>$\mathbf{r} \cdot \mathbf{p} - i$</td></tr><tr><td>$\Gamma_4$</td><td>$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r)$</td><td>$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Gamma_5$</td><td>$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r)$</td><td>$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Theta$</td><td>$r(H_S - W)$</td><td>$r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\}$</td></tr><tr><td></td><td>$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$</td><td>$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$</td></tr><tr><td>$A$</td><td>$1/2m$</td><td>$1$</td></tr><tr><td>$B$</td><td>$-W$</td><td>$m^2 - E^2$</td></tr><tr><td>$C$</td><td>$-\alpha$</td><td>$-2\alpha E$</td></tr></table>	H	$\frac{p^2}{2m} - \frac{\alpha}{r}$	$\sqrt{p^2 + m^2} - \frac{\alpha}{r}$	A_4	$\mathbf{r} \cdot \mathbf{p} - i$	$\mathbf{r} \cdot \mathbf{p} - i$	Γ_4	$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r)$	$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right)$	Γ_5	$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r)$	$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right)$	Θ	$r(H_S - W)$	$r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\}$		$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$	$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$	A	$1/2m$	1	B	$-W$	$m^2 - E^2$	C	$-\alpha$	$-2\alpha E$	<table><tr><th>H</th><th>$\frac{p^2}{2m} - \frac{\alpha}{r}$</th><th>$\sqrt{p^2 + m^2} - \frac{\alpha}{r}$</th></tr><tr><td>$A_4$</td><td>$\mathbf{r} \cdot \mathbf{p} - i$</td><td>$\mathbf{r} \cdot \mathbf{p} - i$</td></tr><tr><td>$\Gamma_4$</td><td>$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r)$</td><td>$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Gamma_5$</td><td>$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r)$</td><td>$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right)$</td></tr><tr><td>$\Theta$</td><td>$r(H_S - W)$</td><td>$r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\}$</td></tr><tr><td></td><td>$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$</td><td>$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$</td></tr><tr><td>$A$</td><td>$1/2m$</td><td>$1$</td></tr><tr><td>$B$</td><td>$-W$</td><td>$m^2 - E^2$</td></tr><tr><td>$C$</td><td>$-\alpha$</td><td>$-2\alpha E$</td></tr></table>	H	$\frac{p^2}{2m} - \frac{\alpha}{r}$	$\sqrt{p^2 + m^2} - \frac{\alpha}{r}$	A_4	$\mathbf{r} \cdot \mathbf{p} - i$	$\mathbf{r} \cdot \mathbf{p} - i$	Γ_4	$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} - r)$	$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} - r - \frac{\alpha^2}{r}\right)$	Γ_5	$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r)$	$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right)$	Θ	$r(H_S - W)$	$r\left\{\left(H_{KG} + \frac{\alpha}{r}\right)^2 - \left(E + \frac{\alpha}{r}\right)^2\right\}$		$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$	$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$	A	$1/2m$	1	B	$-W$	$m^2 - E^2$	C	$-\alpha$	$-2\alpha E$
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Γ_5	$\frac{1}{2}(r\mathbf{p} \cdot \mathbf{p} + r)$	$\frac{1}{2}\left(r\mathbf{p} \cdot \mathbf{p} + r - \frac{\alpha^2}{r}\right)$																																																									
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	$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$	$A(\Gamma_5 + \Gamma_4) + B(\Gamma_5 - \Gamma_4) + C$																																																									
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B	$-W$	$m^2 - E^2$																																																									
C	$-\alpha$	$-2\alpha E$																																																									

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gilmore-lie-groups_ TAB_029	267	■0.953	$\begin{tabular}[t]{ l l }\hline Particle & Rest energy (MeV) \\ \hline electron e^\pm & 0.511 \\ \hline mu meson \mu^\pm & 105.7 \\ \hline tau meson \tau^\pm & 1784.0 \\ \hline proton, antiproton p^\pm & 938.26 \\ \hline deuteron d^+ & 1875.6 \\ \hline tritium t^+ & 2809.4 \\ \hline pi meson \pi^\pm & 139.6 \\ \hline sigma meson \Sigma^\pm & 1385.0 \\ \hline cascade meson \Xi^- & 1533.0 \\ \hline omega \Omega^- & 1672.0 \\ \hline\end{tabular}$	<table><tr><th>Particle</th><th>Rest energy (MeV)</th></tr><tr><td>electron $e^\pm$</td><td>0.511</td></tr><tr><td>mu meson $\mu^\pm$</td><td>105.7</td></tr><tr><td>tau meson $\tau^\pm$</td><td>1784.0</td></tr><tr><td>proton, antiproton $p^\pm$</td><td>938.26</td></tr><tr><td>deuteron d^+</td><td>1875.6</td></tr><tr><td>tritium t^+</td><td>2809.4</td></tr><tr><td>pi meson $\pi^\pm$</td><td>139.6</td></tr><tr><td>sigma meson $\Sigma^\pm$</td><td>1385.0</td></tr><tr><td>cascade meson Ξ^-</td><td>1533.0</td></tr><tr><td>omega Ω^-</td><td>1672.0</td></tr></table>	Particle	Rest energy (MeV)	electron $e^\pm$	0.511	mu meson $\mu^\pm$	105.7	tau meson $\tau^\pm$	1784.0	proton, antiproton $p^\pm$	938.26	deuteron d^+	1875.6	tritium t^+	2809.4	pi meson $\pi^\pm$	139.6	sigma meson $\Sigma^\pm$	1385.0	cascade meson Ξ^-	1533.0	omega Ω^-	1672.0	<table><tr><th>Particle</th><th>Rest energy (MeV)</th></tr><tr><td>electron $e^\pm$</td><td>0.511</td></tr><tr><td>mu meson $\mu^\pm$</td><td>105.7</td></tr><tr><td>tau meson $\tau^\pm$</td><td>1784.0</td></tr><tr><td>proton, antiproton $p^\pm$</td><td>938.26</td></tr><tr><td>deuteron d^+</td><td>1875.6</td></tr><tr><td>tritium t^+</td><td>2809.4</td></tr><tr><td>pi meson $\pi^\pm$</td><td>139.6</td></tr><tr><td>sigma meson $\Sigma^\pm$</td><td>1385.0</td></tr><tr><td>cascade meson Ξ^-</td><td>1533.0</td></tr><tr><td>omega Ω^-</td><td>1672.0</td></tr></table>	Particle	Rest energy (MeV)	electron $e^\pm$	0.511	mu meson $\mu^\pm$	105.7	tau meson $\tau^\pm$	1784.0	proton, antiproton $p^\pm$	938.26	deuteron d^+	1875.6	tritium t^+	2809.4	pi meson $\pi^\pm$	139.6	sigma meson $\Sigma^\pm$	1385.0	cascade meson Ξ^-	1533.0	omega Ω^-	1672.0
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gilmore-lie-groups_ TAB_030	269	■1.000	$\begin{tabular}[t]{ l l l l l l l l }\hline Mercury & Venus & Earth & Mars & Jupiter & Saturn & Uranus & Neptune \\ \hline 0.39 & 0.72 & 1.00 & 1.52 & 5.20 & 9.54 & 19.18 & 30.06 \\ \hline\end{tabular}$	<table><tr><th>Mercury</th><th>Venus</th><th>Earth</th><th>Mars</th><th>Jupiter</th><th>Saturn</th><th>Uranus</th><th>Neptune</th></tr><tr><td>0.39</td><td>0.72</td><td>1.00</td><td>1.52</td><td>5.20</td><td>9.54</td><td>19.18</td><td>30.06</td></tr></table>	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune	0.39	0.72	1.00	1.52	5.20	9.54	19.18	30.06	<table><tr><th>Mercury</th><th>Venus</th><th>Earth</th><th>Mars</th><th>Jupiter</th><th>Saturn</th><th>Uranus</th><th>Neptune</th></tr><tr><td>0.39</td><td>0.72</td><td>1.00</td><td>1.52</td><td>5.20</td><td>9.54</td><td>19.18</td><td>30.06</td></tr></table>	Mercury	Venus	Earth	Mars	Jupiter	Saturn	Uranus	Neptune	0.39	0.72	1.00	1.52	5.20	9.54	19.18	30.06												
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gilmore-lie-groups_ TAB_031	274	■1.000	$\begin{tabular}[t]{ l l l l }\hline Time period & Approach & Strengths & Weaknesses \\ \hline Nineteenth century & Manifestly covariant & Fields have elegant transformation properties & Many fields represent nonphysical states \\ \hline Twentieth century & Hilbert space & All linear superpositions represent physical states & Transformation properties are complicated \\ \hline\end{tabular}$	<table><tr><th>Time period</th><th>Approach</th><th>Strengths</th><th>Weaknesses</th></tr><tr><td>Nineteenth century</td><td>Manifestly covariant</td><td>Fields have elegant transformation properties</td><td>Many fields represent nonphysical states</td></tr><tr><td>Twentieth century</td><td>Hilbert space</td><td>All linear superpositions represent physical states</td><td>Transformation properties are complicated</td></tr></table>	Time period	Approach	Strengths	Weaknesses	Nineteenth century	Manifestly covariant	Fields have elegant transformation properties	Many fields represent nonphysical states	Twentieth century	Hilbert space	All linear superpositions represent physical states	Transformation properties are complicated	<table><tr><th>Time period</th><th>Approach</th><th>Strengths</th><th>Weaknesses</th></tr><tr><td>Nineteenth century</td><td>Manifestly covariant</td><td>Fields have elegant transformation properties</td><td>Many fields represent nonphysical states</td></tr><tr><td>Twentieth century</td><td>Hilbert space</td><td>All linear superpositions represent physical states</td><td>Transformation properties are complicated</td></tr></table>	Time period	Approach	Strengths	Weaknesses	Nineteenth century	Manifestly covariant	Fields have elegant transformation properties	Many fields represent nonphysical states	Twentieth century	Hilbert space	All linear superpositions represent physical states	Transformation properties are complicated																				
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gilmore-lie-groups_ TAB_032	280	■0.815	$\begin{tabular}[t]{ l l l }\hline & k\rangle & \left. \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle \\ \hline \{I, a\} & k\rangle e^{ik\cdot a} & \left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle_{vv';\mu\mu'} \\ \hline \{\Lambda, 0\} & \Lambda k\rangle & \left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(\Lambda) \\ \hline\end{tabular}$	<table><tr><th></th><th>$k\rangle$</th><th>$\left. \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle$</th></tr><tr><td>$\{I, a\}$</td><td>$k\rangle e^{ik\cdot a}$</td><td>$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle_{vv';\mu\mu'}$</td></tr><tr><td>$\{\Lambda, 0\}$</td><td>$\Lambda k\rangle$</td><td>$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(\Lambda)$</td></tr></table>		$ k\rangle$	$\left. \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle$	$\{I, a\}$	$ k\rangle e^{ik\cdot a}$	$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle_{vv';\mu\mu'}$	$\{\Lambda, 0\}$	$ \Lambda k\rangle$	$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(\Lambda)$	<table><tr><th></th><th>$k\rangle$</th><th>$\left. \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle$</th></tr><tr><td>$\{I, a\}$</td><td>$k\rangle e^{ik\cdot a}$</td><td>$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle_{vv';\mu\mu'}$</td></tr><tr><td>$\{\Lambda, 0\}$</td><td>$\Lambda k\rangle$</td><td>$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(\Lambda)$</td></tr></table>		$ k\rangle$	$\left. \begin{array}{c} j \\ \mu \end{array} \begin{array}{c} j' \\ \mu' \end{array} \right\rangle$	$\{I, a\}$	$ k\rangle e^{ik\cdot a}$	$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle_{vv';\mu\mu'}$	$\{\Lambda, 0\}$	$ \Lambda k\rangle$	$\left. \begin{array}{c} j \\ v \end{array} \begin{array}{c} j' \\ v' \end{array} \right\rangle D_{vv';\mu\mu'}^{jj'}(\Lambda)$																										
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gilmore-lie-groups_ TAB_033	290	■ 1.000	<pre>\begin{tabular}[t]{l l l l } \hline & Electric & Magnetic \\ \hline Charge density & \(\rho_e(\mathbf{x}, t)=\sum_j e_j \mathbf{x}_j(t)\) & \(\rho_m(\mathbf{x}, t)=\sum_j m_j \mathbf{x}_j(t)\) \\ \hline Current density & \(\mathbf{J}_e(\mathbf{x}, t)=\sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}\) & \(\mathbf{J}_m(\mathbf{x}, t)=\sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}\) \\ \hline Conservation law & \(\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0\) & \(\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0\) \\ \hline \end{tabular}</pre>	<table><tr><th></th><th>Electric</th><th>Magnetic</th></tr><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$</td></tr></table>		Electric	Magnetic	Charge density	$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$	<table><tr><th colspan="2">Electric</th><th>Magnetic</th></tr><tr><td>Charge density</td><td>$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$</td><td>$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$</td></tr><tr><td>Current density</td><td>$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$</td><td>$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$</td></tr><tr><td>Conservation law</td><td>$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$</td><td>$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$</td></tr></table>	Electric		Magnetic	Charge density	$\rho_e(\mathbf{x}, t) = \sum_j e_j \mathbf{x}_j(t)$	$\rho_m(\mathbf{x}, t) = \sum_j m_j \mathbf{x}_j(t)$	Current density	$\mathbf{J}_e(\mathbf{x}, t) = \sum_j e_j \frac{d\mathbf{x}_j(t)}{dt}$	$\mathbf{J}_m(\mathbf{x}, t) = \sum_j m_j \frac{d\mathbf{x}_j(t)}{dt}$	Conservation law	$\nabla \cdot \mathbf{J}_e(\mathbf{x}, t) + \frac{\partial \rho_e(\mathbf{x}, t)}{\partial t} = 0$	$\nabla \cdot \mathbf{J}_m(\mathbf{x}, t) + \frac{\partial \rho_m(\mathbf{x}, t)}{\partial t} = 0$
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gilmore-lie-groups_TAB_034	311	0.852	$\begin{tabular}{ t { l l l l l l }} \hline \multicolumn{3}{ c }{\text{Infinitesimal generators}} \& \multicolumn{3}{ c }{\text{Canonical coordinates}} \& \text{Constraint} \\ \hline \xi(x,y) \& \eta(x,y) \& \zeta(x,y,p) \& R(x,y) \& S(x,y) \& T(x,y,p) \& dS/dR \\ \hline 0 \& 1 \& 0 \& x \& y \& p \& T \\ 1 \& 0 \& 0 \& y \& x \& p \& 1/T \\ 1/a \& -1/b \& 0 \& ax+by \& bx-ay \& p \& (b-aT)/(a+bT) \\ x \& 0 \& -p \& x \& \ln x \& xp \& 1/T \\ 0 \& y \& p \& y \& \ln y \& p/y \& T \\ x/a \& y/b \& (1/b-1/a)p \& y^b/x^a \& b \ln y \& p/x^{(a/b-1)} \& (bT/R)/(bT-aR^{(1/b)}) \\ x \& y \& 0 \& y/x \& \ln y \& p \& (T/R)/(T-R) \\ x \& -y \& -2p \& xy \& \ln y \& x^2p \& (T/R)/(T+R) \\ 2x \& y \& -p \& y^2/x \& \ln y \& yp \& (T/R)/(2T-R) \\ x \& 2y \& p \& y/x^2 \& \ln y \& p/x \& (T/R)/(T-2R) \\ y \& 0 \& -p^2 \& y \& x/y \& x-y/p \& -T/R^2 \\ 0 \& x \& 1 \& x \& y/x \& xp-y \& T/R^2 \\ -y \& x \& 1+p^2 \& \sqrt{x^2+y^2} \& \tan^{-1}(y/x) \& (y-xp)/(x+yp) \& -T/R \\ 1 \& y/x \& (px-y)/x^2 \& y/x \& x \& (xp-y)/x^2 \& 1/T \\ a \& x \& 1 \& x^2-2ay \& x/a \& x-ap \& 1/(2aT) \\ a \& y \& p \& x-a \ln y \& x/a \& p/y \& (1/a)/(1-aT) \\ x \& b \& -p \& e^y/x^b \& y/b \& e^y * p^b \& (bR)^{-1}/[1-b(R/T)^{(1/b)}] \\ \hline y \& b \& -p^2 \& y^2-2bx \& y/b \& y-b/p \& 1/(2bT) \\ 0 \& e^{f(x)} \& f'e^f \& x \& y/e^f \& p-yf' \& T/e^{f(R)} \\ x^2 \& xy \& y-xp \& y/x \& 1/x \& xp-y \& 1/T \\ xy \& y^2 \& yp-xp^2 \& y/x \& 1/y \& y/p-x \& 1/(TR^2) \\ xy \& 0 \& -yp-xp^2 \& y \& (\ln x)/y \& y/(xp)-\ln x \& T/R^2 \\ 0 \& xy \& y+xp \& x \& (\ln y)/x \& xp/y-\ln y \& T/R^2 \\ g(y) \& 0 \& -g'p^2 \& y \& x/g \& 1/p-xg'/g \& T/g(R) \\ 0 \& f(x) \& f' \& x \& y/f \& fp-f'y \& T/f^2(R) \\ f(x) \& 0 \& -f'p \& y \& F \text{ (} F'f=1 \text{)} \& pf \& 1/T \\ 0 \& g(y) \& g'p \& x \& G \text{ (} G'g=1 \text{)} \& p/g \& T \\ x^{k+1} \& kx^ky \& x^k(k^2y/x-p) \& y/x^k \& 1/x^k \& xp-ky \& -k/T \\ kxy^k \& y^{k+1} \& y^k(p-k^2xp^2/y) \& x/y^k \& 1/y^k \& y/p-kx \& -k/T \\ \hline \end{tabular}$	<table><tr><th colspan="2">Infinitesimal generators</th><th colspan="2">Canonical coordinates</th><th>Constraint</th></tr><tr><td>$\xi(x,y)$</td><td>$\eta(x,y)$</td><td>$R(x,y)$</td><td>$S(x,y)$</td><td>$T(x,y,p)$</td><td>dS/dR</td></tr><tr><td>0</td><td>1</td><td>0</td><td>x</td><td>y</td><td>p</td><td>T</td></tr><tr><td>1</td><td>0</td><td>0</td><td>y</td><td>x</td><td>p</td><td>$1/T$</td></tr><tr><td>$1/a$</td><td>$-1/b$</td><td>0</td><td>$ax+by$</td><td>$bx-ay$</td><td>p</td><td>$(b-aT)/(a+bT)$</td></tr><tr><td>x</td><td>0</td><td>$-p$</td><td>y</td><td>$\ln x$</td><td>xp</td><td>$1/T$</td></tr><tr><td>0</td><td>y</td><td>p</td><td>x</td><td>$\ln y$</td><td>p/y</td><td>T</td></tr><tr><td>x/a</td><td>y/b</td><td>$(1/b-1/a)p$</td><td>y^b/x^a</td><td>$b \ln y$</td><td>$p/x^{(a/b-1)}$</td><td>$(bT/R)/(bT-aR^{(1/b)})$</td></tr><tr><td>x</td><td>y</td><td>0</td><td>y/x</td><td>$\ln 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y$	p/y	T	x/a	y/b	$(1/b-1/a)p$	y^b/x^a	$b \ln y$	$p/x^{(a/b-1)}$	$(bT/R)/(bT-aR^{(1/b)})$	x	y	0	y/x	$\ln y$	p	$(T/R)/(T-R)$	x	$-y$	$-2p$	xy	$\ln y$	x^2p	$(T/R)/(T+R)$	$2x$	y	$-p$	y^2/x	$\ln y$	yp	$(T/R)/(2T-R)$	x	$2y$	p	y/x^2	$\ln y$	p/x	$(T/R)/(T-2R)$	y	0	$-p^2$	y	x/y	$x-y/p$	$-T/R^2$	0	x	1	x	y/x	$xp-y$	T/R^2	$-y$	x	$1+p^2$	$\sqrt{x^2+y^2}$	$\tan^{-1}(y/x)$	$(y-xp)/(x+yp)$	$-T/R$	1	y/x	$(px-y)/x^2$	y/x	x	$(xp-y)/x^2$	$1/T$	a	x	1	x^2-2ay	x/a	$x-ap$	$1/(2aT)$	a	y	p	$x-a \ln y$	x/a	p/y	$(1/a)/(1-aT)$	x	b	$-p$	e^y/x^b	y/b	$e^y * p^b$	$(bR)^{-1}/[1-b(R/T)^{(1/b)}]$								y	b	$-p^2$	y^2-2bx	y/b	$y-b/p$	$1/(2bT)$	0	$e^{f(x)}$	$f'e^f$	x	y/e^f	$p-yf'$	$T/e^{f(R)}$	x^2	xy	$y-xp$	y/x	$1/x$	$xp-y$	$1/T$	xy	y^2	$yp-xp^2$	y/x	$1/y$	$y/p-x$	$1/(TR^2)$	xy	0	$-yp-xp^2$	y	$(\ln x)/y$	$y/(xp)-\ln x$	T/R^2	0	xy	$y+xp$	x	$(\ln y)/x$	$xp/y-\ln y$	T/R^2	$g(y)$	0	$-g'p^2$	y	x/g	$1/p-xg'/g$	$T/g(R)$	0	$f(x)$	f'	x	y/f	$fp-f'y$	$T/f^2(R)$	$f(x)$	0	$-f'p$	y	$F \text{ (} F'f=1 \text{)}$	pf	$1/T$	0	$g(y)$	$g'p$	x	$G \text{ (} G'g=1 \text{)}$	p/g	T	x^{k+1}	kx^ky	$x^k(k^2y/x-p)$	y/x^k	$1/x^k$	$xp-ky$	$-k/T$	kxy^k	y^{k+1}	$y^k(p-k^2xp^2/y)$	x/y^k	$1/y^k$	$y/p-kx$	$-k/T$
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gilmore-lie-groups_ TAB_037	319	■1.000	$\begin{tabular}{t}{ l l l } \hline v_i & \text{Infinitesimal} & e^{\epsilon v_i} f(x, t) = \\ \hline v_1 & \partial_x & f(x - \epsilon, t) \\ \hline v_2 & \partial_t & f(x, t - \epsilon) \\ \hline v_3 & u \partial_u & e^\epsilon f(x, t) \\ \hline v_4 & x \partial_x + 2t \partial_t & f(e^{-\epsilon} x, e^{-2\epsilon} t) \\ \hline v_5 & 2t \partial_x - x u \partial_u & e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t) \\ \hline v_6 & 4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u & \lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t) \text{ where } \lambda^2 = 1/(1 + 4\epsilon t) \\ \hline \end{tabular}$	<table><tr><td>v_i</td><td>Infinitesimal</td><td>$e^{\epsilon v_i} f(x, t) =$</td></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t) \text{ where } \lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t) \text{ where } \lambda^2 = 1/(1 + 4\epsilon t)$	<table><tr><td>v_i</td><td>Infinitesimal</td><td>$e^{\epsilon v_i} f(x, t) =$</td></tr><tr><td>$v_1$</td><td>$\partial_x$</td><td>$f(x - \epsilon, t)$</td></tr><tr><td>$v_2$</td><td>$\partial_t$</td><td>$f(x, t - \epsilon)$</td></tr><tr><td>$v_3$</td><td>$u \partial_u$</td><td>$e^\epsilon f(x, t)$</td></tr><tr><td>$v_4$</td><td>$x \partial_x + 2t \partial_t$</td><td>$f(e^{-\epsilon} x, e^{-2\epsilon} t)$</td></tr><tr><td>$v_5$</td><td>$2t \partial_x - x u \partial_u$</td><td>$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$</td></tr><tr><td>$v_6$</td><td>$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$</td><td>$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$</td></tr></table>	v_i	Infinitesimal	$e^{\epsilon v_i} f(x, t) =$	v_1	∂_x	$f(x - \epsilon, t)$	v_2	∂_t	$f(x, t - \epsilon)$	v_3	$u \partial_u$	$e^\epsilon f(x, t)$	v_4	$x \partial_x + 2t \partial_t$	$f(e^{-\epsilon} x, e^{-2\epsilon} t)$	v_5	$2t \partial_x - x u \partial_u$	$e^{-\epsilon x + \epsilon^2 t} f(x - 2\epsilon t, t)$	v_6	$4xt \partial_x + 4t^2 \partial_t - (x^2 + 2t) u \partial_u$	$\lambda e^{-\epsilon \lambda^2 x^2} f(\lambda^2 x, \lambda^2 t)$ where $\lambda^2 = 1/(1 + 4\epsilon t)$
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gilmore-lie-groups_ TAB_038	320	■1.000	$\begin{tabular}{t}{ l l } \hline \text{Symmetry} & \text{Conserved quantity} \\ \hline \text{Space displacements} & \text{momentum} \\ \hline \text{Time displacements} & \text{energy} \\ \hline \text{Space-time displacements} & \text{four-momentum} \\ \hline \text{Rotations} & \text{angular momentum} \\ \hline \end{tabular}$	<table><tr><td>Symmetry</td><td>Conserved quantity</td></tr><tr><td>Space displacements</td><td>momentum</td></tr><tr><td>Time displacements</td><td>energy</td></tr><tr><td>Space-time displacements</td><td>four-momentum</td></tr><tr><td>Rotations</td><td>angular momentum</td></tr></table>	Symmetry	Conserved quantity	Space displacements	momentum	Time displacements	energy	Space-time displacements	four-momentum	Rotations	angular momentum	<table><tr><td>Symmetry</td><td>Conserved quantity</td></tr><tr><td>Space displacements</td><td>momentum</td></tr><tr><td>Time displacements</td><td>energy</td></tr><tr><td>Space-time displacements</td><td>four-momentum</td></tr><tr><td>Rotations</td><td>angular momentum</td></tr></table>	Symmetry	Conserved quantity	Space displacements	momentum	Time displacements	energy	Space-time displacements	four-momentum	Rotations	angular momentum																						
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